

Mohamed Tarek Abdelhady

Kafr El-Sheikh, Egypt | +20 1090637406 | mohammed.tarek.abdelhady.ali@gmail.com
github.com/mohamed22tarek5 | linkedin.com/in/mohamed-tarek

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with a strong foundation in Embedded Systems, Industrial Automation, PCB Design, Internet of Things (IoT), and Renewable Energy Systems. Skilled in PLC programming using Siemens TIA Portal, classic control systems, embedded C programming, and hardware-software integration through academic projects and hands-on training.

Passionate about automation, control systems, and industrial technologies, with a commitment to continuous learning and solving real-world engineering challenges. Active member of IEEE and Google Developer Student Clubs (GDSC). Seeking a Junior Automation Engineer position where I can apply my technical knowledge, contribute to innovative automation solutions, and grow in a professional engineering environment.

EDUCATION

Bachelor of Engineering in Electronics & Communication Engineering Expected Jul 2026
Higher Institute of Engineering and Technology in Manzala, Egypt

- **Relevant Coursework:** MATLAB/Simulink, PCB Design, Digital Electronics, Communication Systems, Renewable Energy Systems, Control Systems, Microprocessors
- **Grade:** Good

High School Diploma Expected 2021
Rokha Official Language School, Egypt

TECHNICAL SKILLS

- **Automation & Control:** PLC Programming (Siemens TIA Portal, Ladder Logic), Classic Control Systems, Variable Frequency Drives (VFD), Control Systems Design, Industrial Panel Building
- **Embedded Systems:** Arduino Development, Electronic Circuit Design, PCB Design, IoT Systems
- **Software & Tools:** Siemens TIA Portal, Factory I/O, Altium Designer, Proteus, Multisim, CAdE_SIMU, EKTS, Arduino IDE, MATLAB/Simulink, Firebase
- **Programming Languages:** Python, Embedded C, Ladder Logic

PROFESSIONAL EXPERIENCE

HA (Hassan Ali Consulting Group) Aug 2025
Industrial Automation Trainee 140 hours

- Completed comprehensive training in Classic Control, PLC programming, Motor Drives, and Panel Building
- Developed and tested 20+ Siemens TIA Portal PLC programs, halving Factory I/O simulation errors through iterative debugging.
- Configured Variable Frequency Drives (VFDs) for three industrial motor control applications, reducing commissioning time during simulation.

3M Company Aug 2024
Engineering Intern 160 hours

- Supported manufacturing process optimization initiatives that improved production efficiency.
- Collaborated with a multidisciplinary engineering team to support production optimization initiatives.

Petrobel (Belayim Petroleum Company) Jul 2025
Engineering Intern 56 hours

- Assisted engineers in petroleum operations and field activities.
- Participated in safety protocol training and compliance checks
- Contributed to documentation and reporting of operational procedures

PROJECTS

Design, Modeling, Implementation, and IoT-Based Monitoring of a 60-W Photovoltaic Renewable Energy System (Graduation Project, Grade: Excellent)

- Designed and implemented a 60W photovoltaic renewable energy system with integrated IoT monitoring.
- Modeled and simulated PV system performance using MATLAB/Simulink for optimal efficiency analysis
- Implemented a TL494-based buck converter for efficient battery charging regulation
- Developed a conventional inverter using timer 555 for basic AC power conversion
- Developed a pure sine-wave inverter using EGS002 SPWM controller module for clean power output
- Created IoT monitoring board for real-time data collection and analysis using Firebase
- Developed a Firebase-based web dashboard for real-time monitoring of the system.

PLC Programming Projects (Siemens TIA Portal)

- Developed and tested PLC programs for industrial applications including: Color Detection System, Pick-and-Place System, Elevator Control System, Automatic Filling Tank, Traffic Light Control, Cyclic Motor Operation, and Temperature Monitoring System
- Utilized Ladder Logic programming with timers, counters, comparators, and mathematical operations
- Simulated and validated programs using Factory I/O industrial simulation software

Classic Control Projects

- Designed and implemented control circuits including: Sequential Motor Control, Star-Delta Starter with Forward/Reverse, Automatic Washing Machine Controller, and Timed Motor Operations
- Created detailed wiring diagrams and tested circuit functionality under various load conditions
- Developed comprehensive documentation for each control system design

Smart Blind Glasses

- Developed wearable assistive device using ultrasonic sensors for obstacle detection and navigation assistance.
- Implemented dual feedback system using haptic (vibration motor) and auditory (buzzer) signals for user awareness.
- Designed portable, low-power solution with user testing and feedback-driven improvements.

RFID-Based Attendance & Person Counter

- Designed RFID-based system for automated attendance tracking and room occupancy counting.
- Implemented automated lighting control based on occupancy status for energy efficiency.
- Developed user-friendly administrator interface for data visualization and access.

CERTIFICATIONS & TRAINING

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| • Industrial Automation Training
Classic Control, PLC Programming, Motor Drives, Panel Building — HA Consulting Group | 140 hours, Aug 2025 |
| • CCNA (Cisco Certified Network Associate)
Networking fundamentals, routing, switching, and network security | 24 hours, Jul 2024 |
| • Python & Machine Learning
Python programming, data analysis, and machine learning fundamentals | 24 hours, Jun 2025 |
| • Home Automation Training
Smart home systems, IoT integration, and automation protocols | Jul–Aug 2024 |
| • Network Fundamentals L1
Basic networking concepts and protocols | Jul 2023 |

LEADERSHIP & ACTIVITIES

- **IEEE Member (Vice Chair 2025)** — Active participant in technical workshops and engineering community events; led student chapter activities and organized technical seminars; coordinated team projects and mentored junior members
- **Google Developer Student Clubs (GDSC) Head Technical 2025** — Engaged in collaborative technology learning, development initiatives, and hackathons; contributed to open-source projects and community development